

Experiment 1

Footprinting on the Target Network using Search Engines, Web Services, and Social Networking Sites

1. Aim

To perform passive footprinting on a simulated target network (example.edu) using search engines, web services, and social networking sites to gather publicly available organizational information.

Disclaimer: All targets and outputs in this document are simulated using RFC-compliant example domains and IP addresses for educational purposes.

2. Theoretical Background

Footprinting is the first phase of the ethical hacking process. The objective is to collect as much information as possible about the target system or network to identify paths for further analysis. Passive footprinting relies on open-source intelligence (OSINT) gathering without sending direct traffic or probes to the target's servers.

- **Search Engines:** Utilizing advanced operators (dorks) to filter results.
- **Web Services:** Using third-party sites like Netcraft or BuiltWith to analyze web technology stacks.
- **Social Networking:** Gathering organizational structure and employee details via LinkedIn, Twitter, etc.

3. Detailed Procedure

Step 1: Search Engine Dorking

Utilize Google Advanced Search operators to extract hidden directories, login portals, and sensitive documents.

- Search for general indexed pages: `site:example.edu`
- Search for exposed PDF documents (brochures, internal guidelines):
`site:example.edu filetype:pdf`

- Search for administrative login pages: `site:example.edu inurl:admin | inurl:login`
- Search for exposed directory listings: `site:example.edu intitle:"index of"`

Step 2: Analyzing Web Services

Use third-party platforms to identify the technological footprint of the target.

- Navigate to a technology profiler (e.g., BuiltWith or Wappalyzer).
- Enter the target domain (example.edu).
- Review the frameworks, web servers, and scripting languages in use.
- Use `crt.sh` to analyze SSL transparency logs and uncover hidden subdomains.

Step 3: Social Networking OSINT

Explore publicly available social media profiles to map out human resources.

- Search the organization on LinkedIn to find the standard email formatting (e.g., `first.last@example.edu`).
- Identify key personnel (IT Admins, Network Engineers).
- Look for technical job postings which often reveal backend technologies (e.g., "Looking for a Sysadmin with Cisco ASA experience").

4. Simulated Observation & Outputs

Category	Query / Method	Sample Extracted Output
Indexed Pages	<code>site:example.edu</code>	1,240 pages indexed. Main sections: /admissions, /departments, /library.
Exposed Documents	<code>site:example.edu filetype:pdf</code>	Found: Student_Handbook.pdf, Faculty_Contact_List_2023.pdf, IT_Policy.pdf
Admin Portals	<code>site:example.edu inurl:login</code>	Located: portal.example.edu/login.php, staff.example.edu/admin
Tech Stack (Web Service)	BuiltWith Profiler	Server: Apache 2.4.x Language: PHP 7.4, ReactJS Analytics: Google Analytics

Category	Query / Method	Sample Extracted Output
Subdomains (crt.sh)	SSL Certificate Search	dev.example.edu, webmail.example.edu, vpn.example.edu
Social Media (LinkedIn)	Company Profile Search	Identified 15 IT staff members. Email format inferred as initial+lastname@example.edu.

5. Result & Conclusion

Passive footprinting was successfully performed on the target domain using search engines, web services, and social media. The gathered intelligence includes the organization's technology stack, employee structures, and hidden web portals. This information maps out the attack surface without alerting the target organization.